

EDUCATION	Zhejiang University <i>Undergraduate Student</i> • Major: Computer Science and Technology. • Minor Program: Advanced Honor Class for Engineering Education(only 50 science&engineering students selected in each Grade), Chu Kochen Honors College, Zhejiang University. • GPA: 3.99/4.3	Hangzhou, China 2022 - 2027 (<i>expected</i>)
RESEARCH INTERESTS	My current research interests focus on designing novel model architectures, particularly efficient attention mechanisms, and on scaling large language models. Previously, I worked on interpretability and trustworthy machine learning, motivated by a desire to uncover the underlying “physics” of neural networks and to ensure their controllability and reliability. My research direction has shifted from DECONSTRUCTION to CONSTRUCTION, with the belief that “What I cannot create, I do not understand”. The next paradigm in artificial intelligence will not emerge from merely refining existing frameworks, but from building fundamentally new ones. My long-term goal is to develop efficient, scalable models with genuine agency.	
PUBLICATIONS AND PREPRINTS	1. Xunhao Lai, Weiqi Xu, Yufeng Yang, Qiaorui Chen, Yang Xu, Lunbin Zeng, Xiaolong Li, Haohai Sun, Haichao Zhu, Vito Zhang, Jinkai Hu, Jiayao Li, Rui Gao, Zekun Li, Songquan Zhu, Jingkai Zhou, Pengyu Zhao. <i>MiniMax Sparse Attention</i>. arXiv preprint, 2026. PDF: https://github.com/MiniMax-AI/MSA/blob/main/docs/MiniMaxSparseAttention.pdf 2. Yang Xu*, Xuanming Zhang*, Samuel Yeh, Jwala Dhamala, Ousmane Dia, Rahul Gupta, Sharon Li. <i>Simulating and Understanding Deceptive Behaviors in Long-Horizon Interactions</i>. ICLR 2026. 3. Yang Xu, Yi Wang, Hao Wang. <i>Tracking the Feature Dynamics in LLM Training: A Mechanistic Study</i>. arXiv preprint arXiv:2412.17626, 2024. 4. Qihan Ren*, Junpeng Zhang*, Yang Xu, Yue Xin, Dongrui Liu, Quanshi Zhang. <i>Towards the Dynamics of a DNN Learning Symbolic Interactions</i>. Neural Information Processing Systems (NeurIPS), 2024. 5. Xu Cheng*, Lei Cheng*, Zhaoran Peng, Yang Xu, Tian Han, Quanshi Zhang. <i>Layerwise Change of Knowledge in Neural Networks</i>. Proceedings of the 41st International Conference on Machine Learning (ICML), PMLR 235:8038-8059, 2024.	
PROJECTS	Fused RoPE & Co-RoPE (Exploration) • Summary: Triton Implementation of Fused RoPE and Exploration of Contextual Improvement of RoPE(Co-RoPE) • Link: https://github.com/Superposition09m/RoPE-CoRoPE Stanford CS336: Language Modeling from Scratch - A1 • Summary: A modular, from-scratch PyTorch implementation featuring modern LM architectural components: BPE tokenization, MHA, RMSNorm, SwiGLU activations, RoPE, and custom AdamW optimizer. Validated on TinyStories (loss 1.34) and OpenWebText (loss 3.78). • Link: https://github.com/Superposition09m/CS336-Mine(course website: https://stanford-cs336.github.io/spring2025/)	

EXPERIENCES	MiniMax Shanghai, China 2026.02-Now • Internship: Algorithm & Engineering Co-design. • Focus: Pretraining and Scaling of Large Language Models.
	Summer Research Program Beijing, China 2025.07-2025.08 • Mentored by Jingyang Yuan (Researcher at Deepseek, the best paper author of ACL 2025) • Focus: Triton-based kernel optimization; Efficient attention mechanisms (linear and sparse attention)
	UW-Madison Madison, USA 2025.03-2025.11(remote) • Remote Research Intern in the School of Computer, Data & Information Sciences • Advisor: Prof. Sharon Li. • Focus: Evaluation of LLM Trustworthiness, Collaboration with Amazon
	Rutgers University New Jersey, USA 2024.07-2025.02(remote since 2024.9) • Visiting Student and Research Intern in the Department of Computer Science. • Advisor: Prof. Hao Wang. • Focus: Mechanistic Interpretability Study of LLMs.
	Shanghai Jiao Tong University Shanghai, China 2023.05-2024.09(remote) • Remote Research Intern in the John Hopcroft Center for Computer Science, School of electronic information and electrical engineering. • Advisor: Prof. Quanshi Zhang. • Focus: Interpretability of Neural Networks and Deep Learning Theory.
AWARDS AND HONORS	• 2nd Scholarship in Zhejiang University. • Win 1st Prize for twice(2022, 2023) in Zhejiang Division of National Mathematics Competition for College Students.
ENGLISH PROFICIENCY	TOEFL iBT: 102 (Listening: 30, Reading: 28, Speaking: 20, Writing: 24) <i>March 2024</i> Activities: Member of ZJUFLA (Zhejiang University Foreign Language Association), English Corner Organizer for 2 semesters. <i>Sept. 2023 – June 2024</i>
ACADEMIC SERVICES	Reviewer for: <i>International Conference on Learning Representations (ICLR) 2025, North American Chapter of the Association for Computational Linguistics (NAACL) 2025, ICLR 2026.</i>